

Lecture 2: Machine Learning or AI Basics Continued

Scalars, Vectors, and Matrices

Scalars

- **Scalar:** A single real number used to represent magnitude or scale.
 - Example: learning rate $\eta = 0.01$, bias term $b = 2.5$

Vectors

- **Vector:** An ordered collection of numbers representing a data point or direction.
 - Example: $x = [25, 60000, 16]^T$ (age, income, years of education)
- **Vector addition:** Combines two vectors element-wise.
 - Example: $[1, 2] + [3, 4] = [4, 6]$
- **Scalar multiplication:** Scales a vector's magnitude.
 - Example: $2[1, 3] = [2, 6]$
- **Dot product:** Measures similarity or projection between vectors.
 - Example: $[1, 2] \cdot [3, 4] = 11$

Matrices

- **Matrix:** A 2D structure representing datasets or linear transformations.
 - Example: Data matrix $X \in \mathbb{R}^{n \times d}$

Matrix Operations

- **Matrix multiplication:** Applies a linear transformation to data.
 - Example: $y = Wx$ in linear regression
- **Transpose:** Flips rows and columns, enabling dot products and symmetry.
 - Example: $x^T x$ computes squared norm
- **Identity matrix:** Acts as multiplicative neutral element.
 - Example: $Ix = x$

Determinants and Inverses

- **Determinant:** Measures volume scaling and invertibility.
 - Example: $\det = 0 \Rightarrow$ features are linearly dependent
- **Matrix inverse:** Reverses a linear transformation.
 - Example: Solving $Ax = b$ as $x = A^{-1}b$ (conceptually)

Systems of Linear Equations and Solvers

- **Linear system:** Set of equations solved simultaneously.
 - Example: fitting regression weights
- **Gaussian elimination:** Converts system to solvable triangular form.
 - Example: taught as row-reduction in high school
- **LU decomposition:** Splits matrix into lower and upper triangular parts.
 - Example: speeds up repeated solves
- **QR decomposition:** Orthogonalizes features.
 - Example: numerically stable least squares
- **Least squares:** Minimizes squared prediction error.
 - Example: fitting a line through noisy data

Linear System

$$Ax = b$$

Explanation:

- $A \in \mathbb{R}^{n \times n}$: coefficient matrix
- $x \in \mathbb{R}^n$: unknown variables
- $b \in \mathbb{R}^n$: outputs

Gaussian Elimination (Step-by-Step)

Augmented matrix:

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 3 & 6 \end{array} \right]$$

Row operations:

- Swap rows
- Eliminate lower entries
- Back-substitution

LU Decomposition

LU decomposition is used to **solve linear systems efficiently**:

$$A\mathbf{x} = \mathbf{b}$$

where $A \in \mathbb{R}^{n \times n}$ is a square matrix.

Instead of solving this system directly, we factor:

$$A = LU$$

Definition

- L : Lower triangular matrix (zeros above diagonal)
- U : Upper triangular matrix (zeros below diagonal)

So,

$$A = LU$$

Sometimes with pivoting:

$$PA = LU$$

where P is a **permutation matrix**.

Example

Let

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$$

We want:

$$A = LU$$

Assume:

$$L = \begin{bmatrix} 1 & 0 \\ \ell_{21} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

Multiply:

$$LU = \begin{bmatrix} u_{11} & u_{12} \\ \ell_{21}u_{11} & \ell_{21}u_{12} + u_{22} \end{bmatrix}$$

Match with A :

- $u_{11} = 2$
- $u_{12} = 3$
- $\ell_{21} = \frac{4}{2} = 2$
- $u_{22} = 7 - 2 \cdot 3 = 1$

So:

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

Solving a system using LU

Suppose:

$$\mathbf{b} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

1. Solve $L\mathbf{y} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

- $y_1 = 5$
- $2(5) + y_2 = 11 \Rightarrow y_2 = 1$

2. Solve $U\mathbf{x} = \mathbf{y}$:

$$\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

- $x_2 = 1$
- $2x_1 + 3 = 5 \Rightarrow x_1 = 1$

Eigenvalues and Matrix Factorization

- **Eigenvector:** Direction unchanged by transformation.
 - Example: principal direction of variance
- **Eigenvalue:** Amount of stretching along eigenvector.
 - Example: large eigenvalue $\Rightarrow$ important direction
- **SVD:** Universal matrix decomposition.
 - Example: image compression

Eigenvalue Equation

$$Av = \lambda v$$

Explanation:

- v : eigenvector (direction)
- λ : eigenvalue (scaling)

Singular Value Decomposition

$$A = U\Sigma V^T$$

Explanation:

- U : left singular vectors
- Σ : singular values (importance)
- V : right singular vectors

Differential Calculus

Derivative Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- **Derivative:** Rate of change of a function.
 - Example: slope of loss curve
- **Partial derivative:** Change w.r.t. one variable.
 - Example: effect of one weight
- **Gradient:** Direction of steepest ascent.
 - Example: points uphill on loss surface
- **Gradient descent:** Iterative error minimization.
 - Example: training linear regression

Gradient

$$\nabla f(x) = \begin{bmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_d \end{bmatrix}$$

Integral Calculus

- **Indefinite integral:** Antiderivative of a function.
 - Example: reversing differentiation
- **Definite integral:** Accumulates quantity over interval.
 - Example: area under curve
- **Expectation:** Average value of random variable.
 - Example: mean of Gaussian

Probability Theory

- **Random variable:** Numeric outcome of random process.
 - Example: class label
- **Bernoulli distribution:** Binary outcomes.
 - Example: spam vs not spam
- **Gaussian distribution:** Continuous natural variation.
 - Example: sensor noise
- **Variance:** Spread of distribution.
 - Example: uncertainty level

Gaussian Distribution

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Explanation:

- μ : mean
- σ^2 : variance

The **Bernoulli distribution** models a single binary (yes/no, success/failure) random variable.

Definition

Let

$$X \sim \text{Bernoulli}(p)$$

Probability Mass Function (PMF)

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

Explanation of each term

- X : random variable representing a single trial
- x : observed outcome (either 0 or 1)
- p : probability of success, where $0 \leq p \leq 1$
- $1 - p$: probability of failure